

Manufacturer	Customer	Business Transaction	
Steel Heat Treatment GmbH 123 Industrial Park 40210 Düsseldorf DE	Automotive Components Inc 456 Manufacturing Drive Detroit 48220 US	Order	ORD-2024-HT-047
		Delivery	DEL-2024-047 · 4750 kg

Product

Product Name	Carbon Steel Round Bar 42CrMo4	Name (EN 10027-1)	42CrMo4
Batch ID	BATCH-2024-047	Number (EN 10027-2)	1.7225
Delivery Condition	+QT - Quenched and Tempered	Name (AISI/SAE)	4140
		Shape	RoundBar - D 80 mm - Length 6000 mm

Heat Treatment

Process: Quenching and Tempering - Lot: HT-2024-11-15-B47 - Furnace: FURNACE-03 - Date: 2024-11-15

Stage	Temperature	Duration	Cooling	Atmosphere
Austenitizing	850 C	45 min		Air
Quenching	60 C	10 min	Oil	
Tempering	550 C	120 min	Air	

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status
Tensile Strength	Rm	985 MPa	900 MPa	1100 MPa	EN ISO 6892-1	✓
Yield Strength	Re	835 MPa	750 MPa		EN ISO 6892-1	✓
Elongation	A	14.5 %	10.0 %		EN ISO 6892-1	✓
Rockwell Hardness	HRC	33 HRC	28 HRC	35 HRC	ASTM E18	✓

Validation

We hereby certify that the material described above has been heat treated and tested in accordance with the requirements of EN 10204:2004 type 3.1 and the specified standards. The results comply with the requirements.

Validated by Dr. Klaus Schmidt, Quality Manager,
Quality Assurance - 2024-11-15